

Circles (Part I): Tangents and Arcs

Introduction to Circles

Example: Circle E is drawn below and abbreviated $\odot E$.

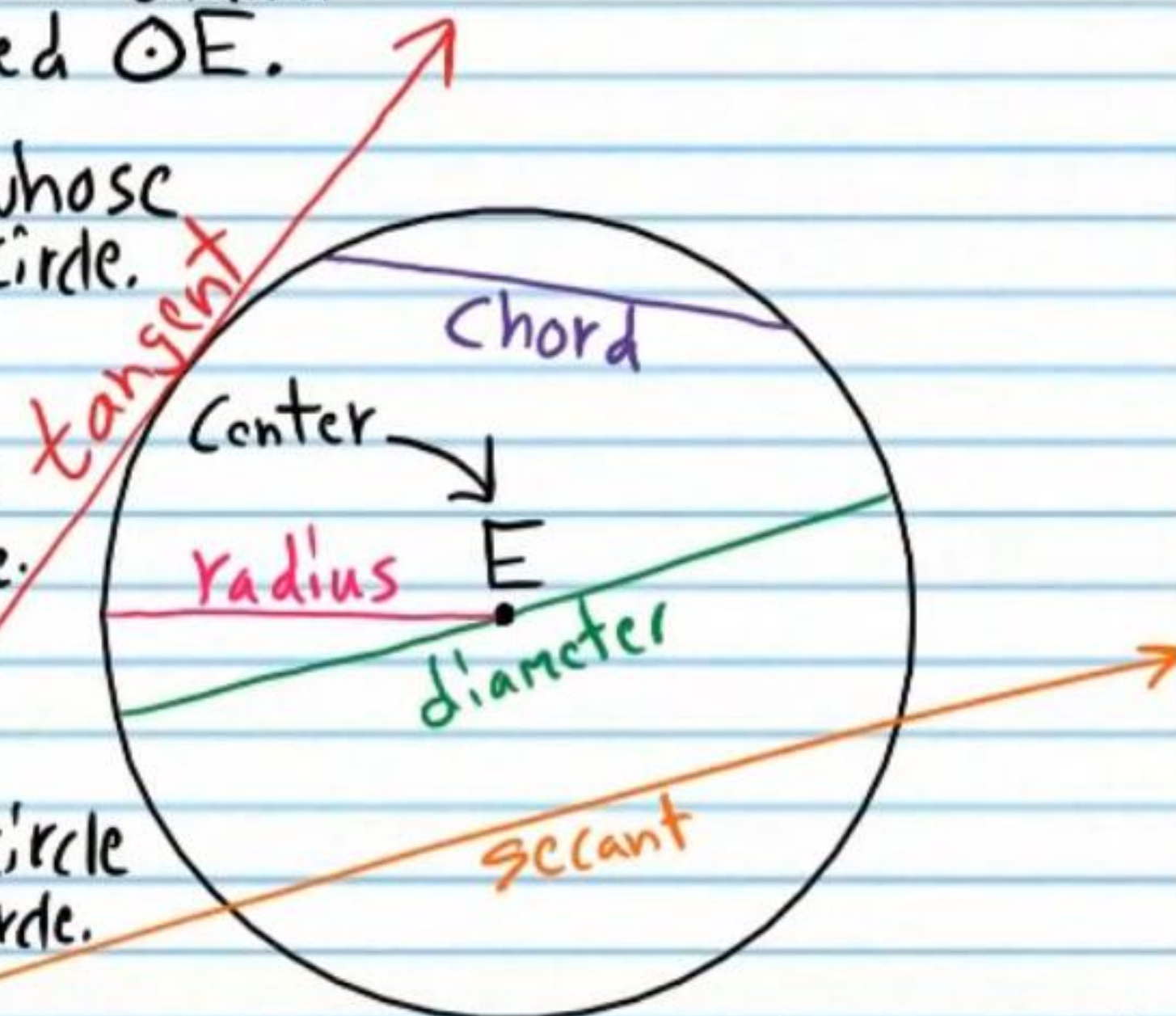
Chord: A segment whose endpoints are on a circle.

Diameter: A chord that passes through the center of a circle.

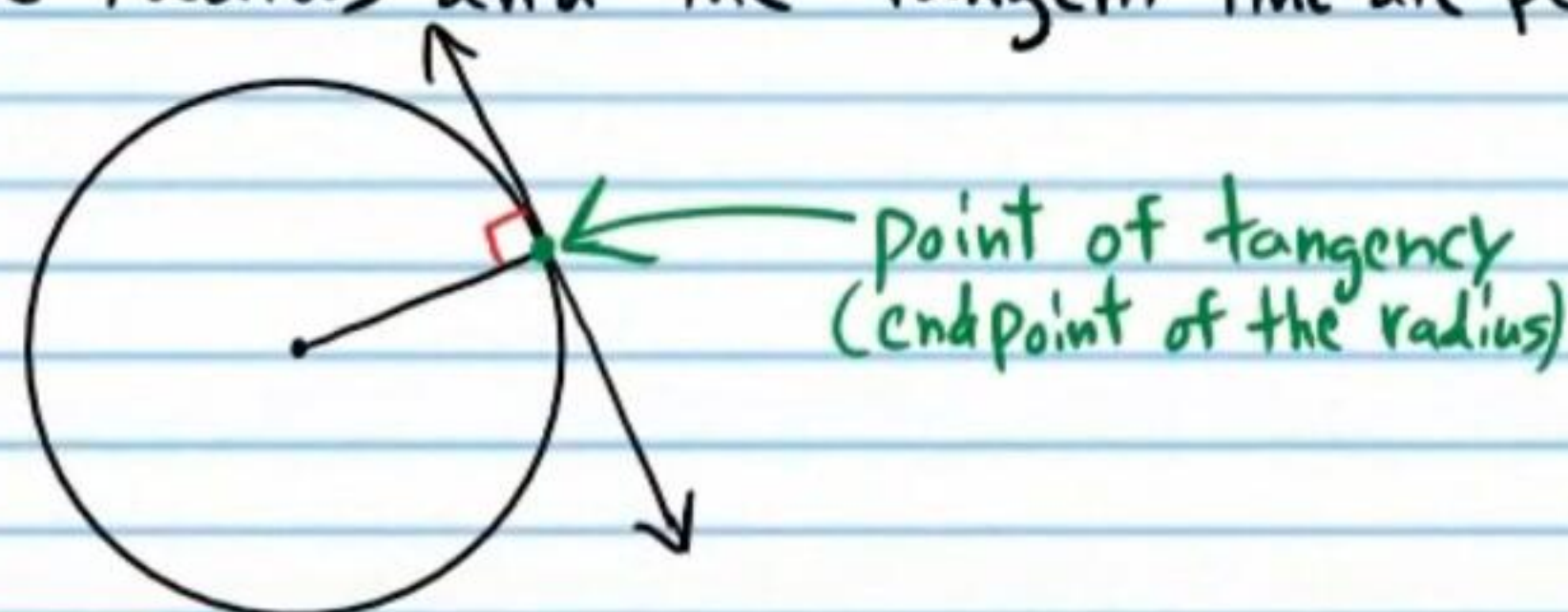
Radius: A segment whose endpoints are on the center of the circle and any point on the circle.

Secant: A line that intersects a circle at two points.

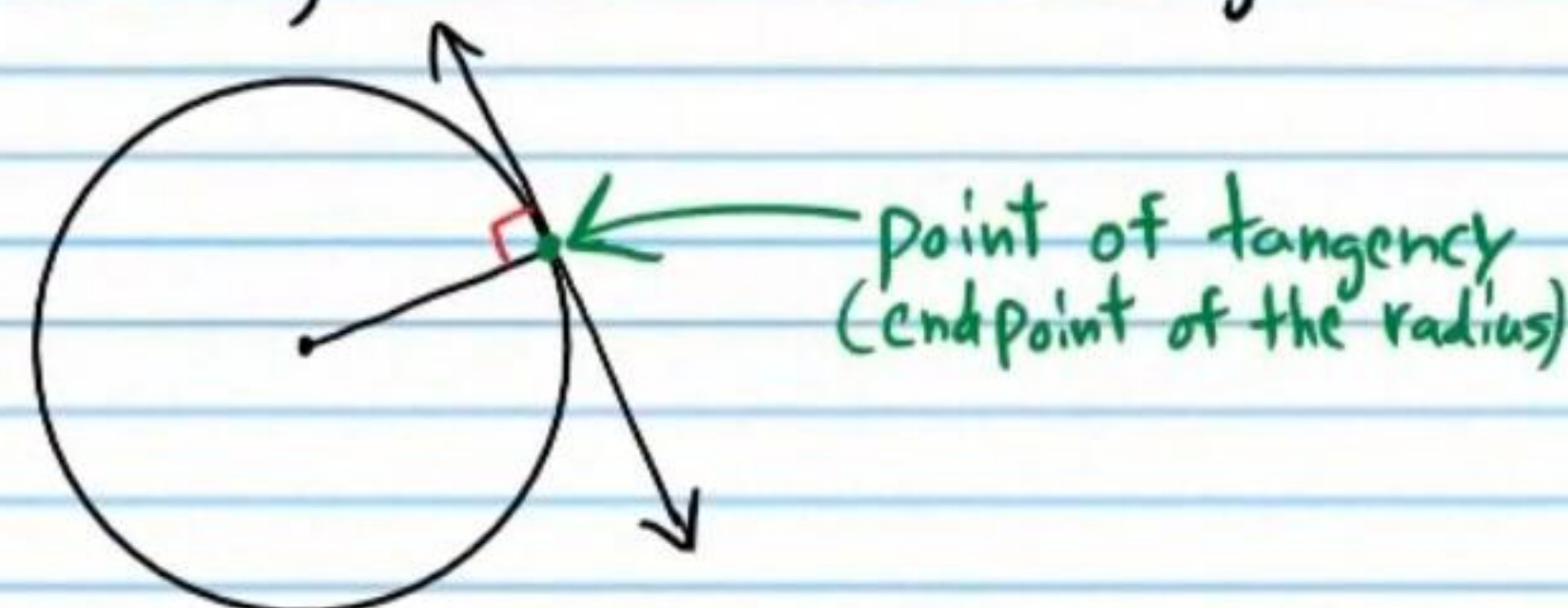
Tangent: A line that intersects a circle at exactly one point.



Perpendicular Radius-Tangent Theorem: If a radius extends to the point of tangency of a tangent line, then the radius and the tangent line are perpendicular.



Perpendicular Radius-Tangent Converse Theorem: If a line is perpendicular to a radius at the endpoint on the circle, then the line is tangent to the circle.

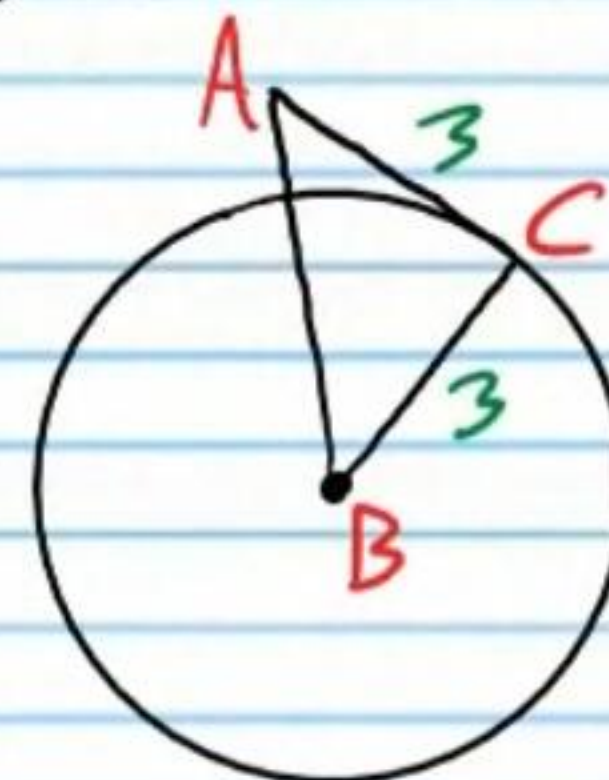
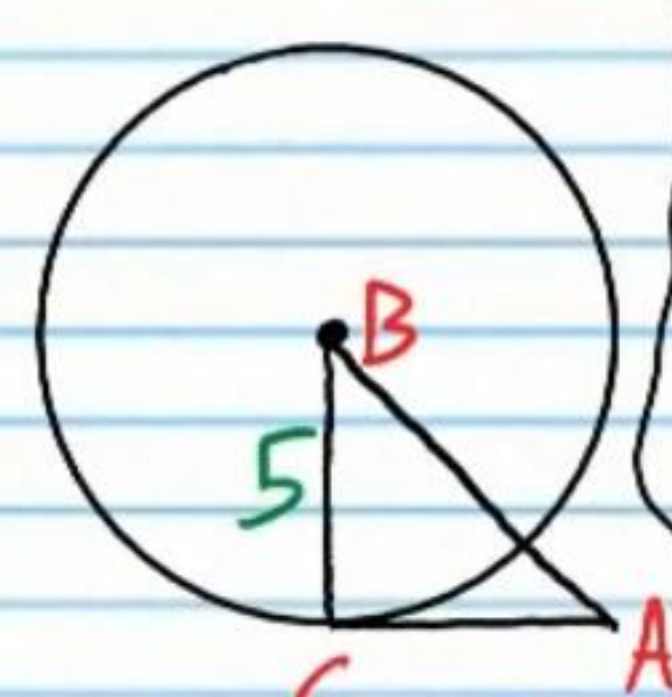
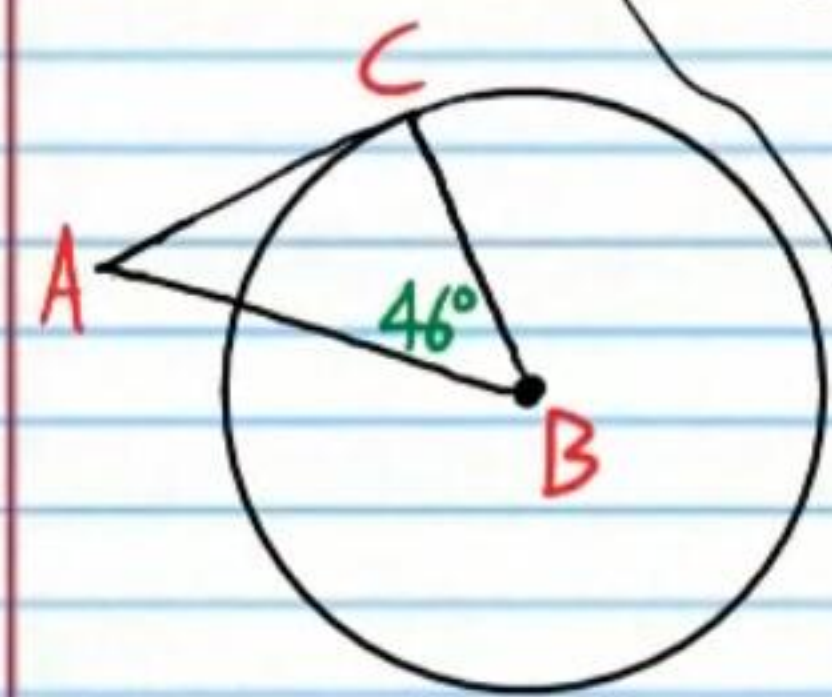


If the line containing $\overline{AC}$ is tangent to $\odot B$ at C, find the following values.

① $m\angle A$

② AC if $AB = \sqrt{61}$

③ AB & $m\angle B$.



$$m\angle A + m\angle B + m\angle C = 180^\circ$$

$$m\angle A + 46 + 90 = 180$$

$$\boxed{m\angle A = 44^\circ}$$

$$a^2 + b^2 = c^2$$

$$5^2 + b^2 = (\sqrt{61})^2$$

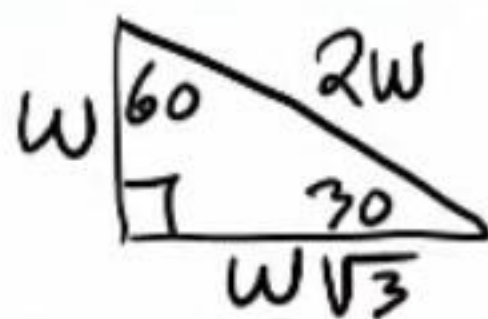
$$25 + b^2 = 61$$

$$b^2 = 36$$

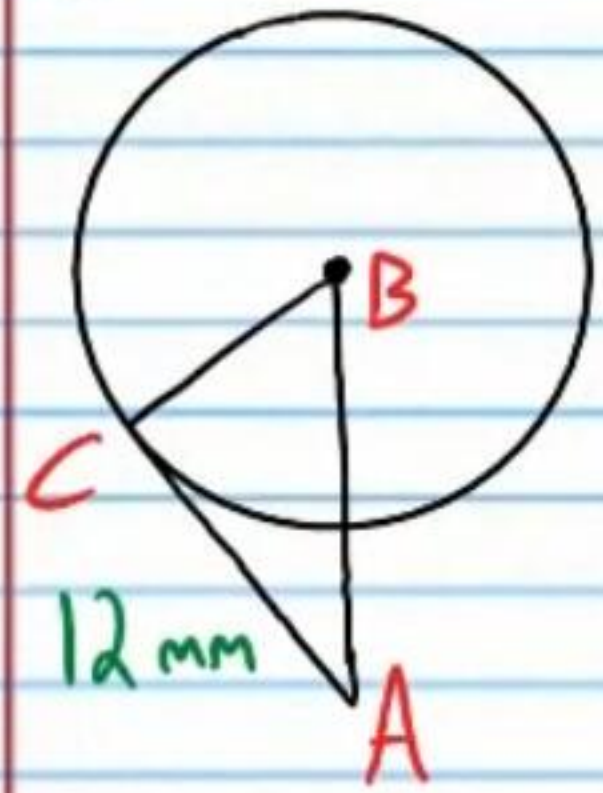
$$\boxed{AC = 6}$$

$$\boxed{AB = 3\sqrt{2}}$$

$$\boxed{m\angle B = 45^\circ}$$



④ CB if $AB = 15\text{ mm}$



$$a^2 + b^2 = c^2$$

$$a^2 + 12^2 = 15^2$$

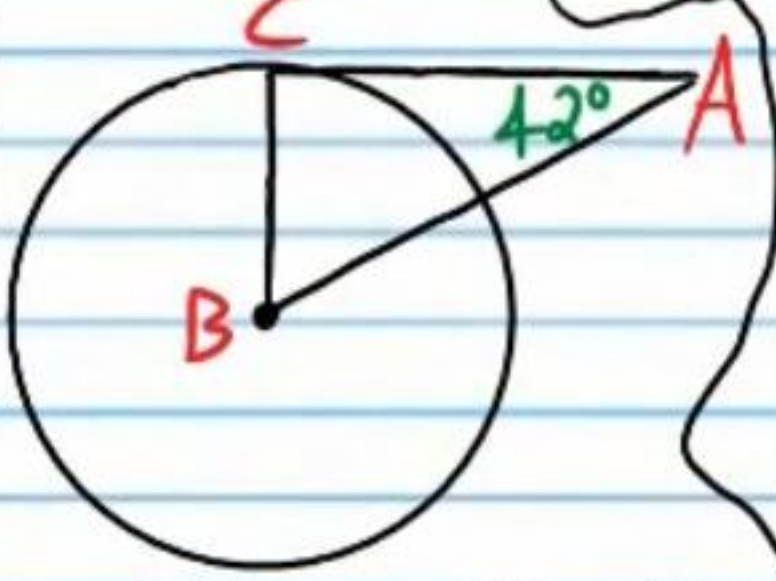
$$a^2 + 144 = 225$$

$$a^2 = 81$$

$$a = 9$$

$$\boxed{CB = 9\text{ mm}}$$

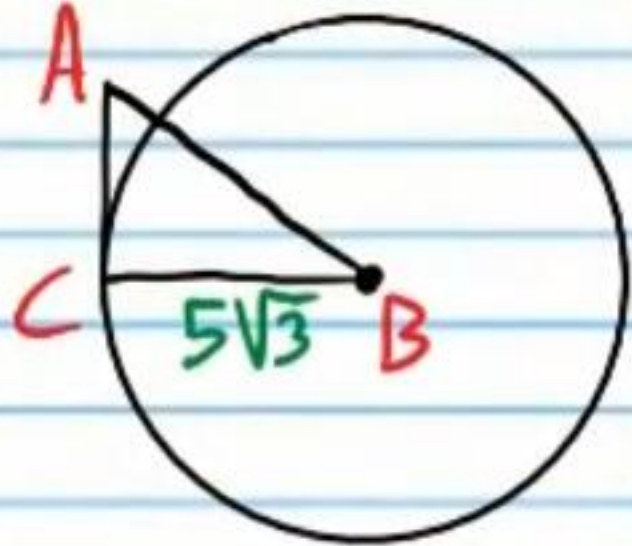
⑤ $m\angle B$



$$90 + 42 + m\angle B = 180$$

$$\boxed{m\angle B = 48^\circ}$$

⑥ $m\angle B$, AC , & AB if $m\angle A = 60^\circ$



$$\boxed{m\angle B = 30^\circ}$$

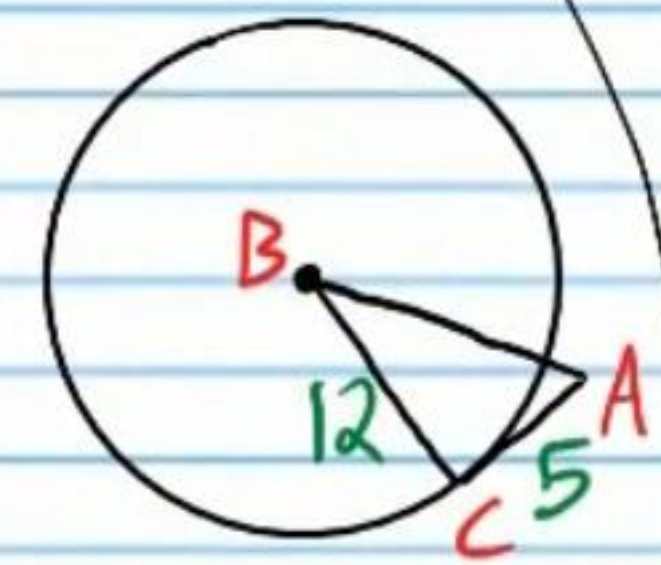
$$\boxed{AC = 5}$$

$$AB = 2(5)$$

$$\boxed{AB = 10}$$

If $\overline{BC}$ is a radius, is the line containing $\overline{AC}$ tangent to $\odot B$? Justify your answer.

⑦ $AB = 13$



$$a^2 + b^2 ? c^2$$

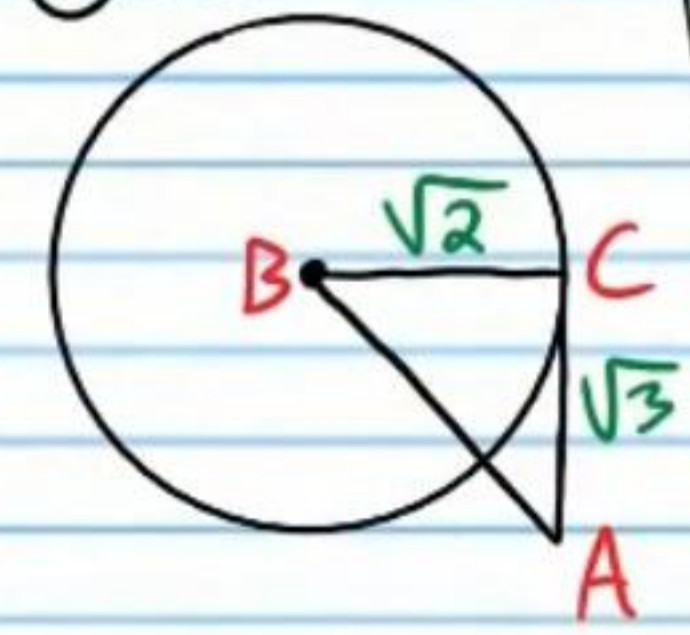
$$12^2 + 5^2 ? 13^2$$

$$144 + 25 ? 169$$

$$169 = 169$$

$$\boxed{\text{Yes}}$$

⑧ $AB = 2.5$



$$a^2 + b^2 ? c^2$$

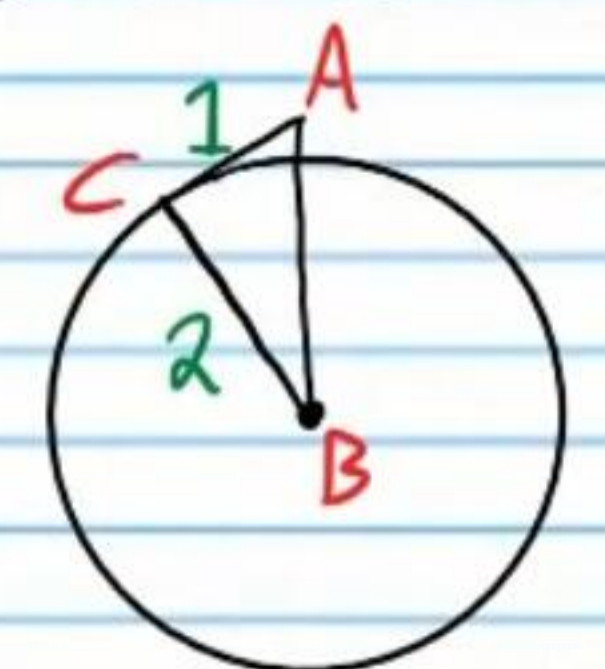
$$(\sqrt{2})^2 + (\sqrt{3})^2 ? (2.5)^2$$

$$2 + 3 ? 6.25$$

$$5 \neq 6.25$$

$$\boxed{\text{No}}$$

⑨ $AB = \sqrt{5}$



$$a^2 + b^2 ? c^2$$

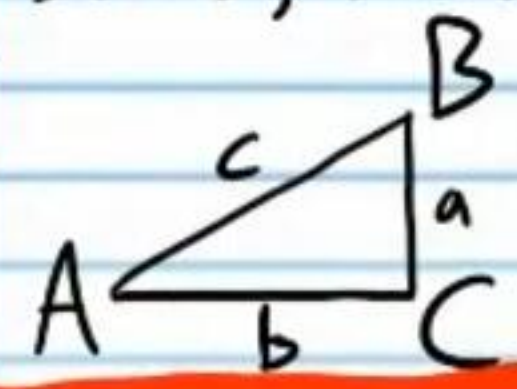
$$2^2 + 1^2 ? (\sqrt{5})^2$$

$$4 + 1 ? 5$$

$$5 = 5$$

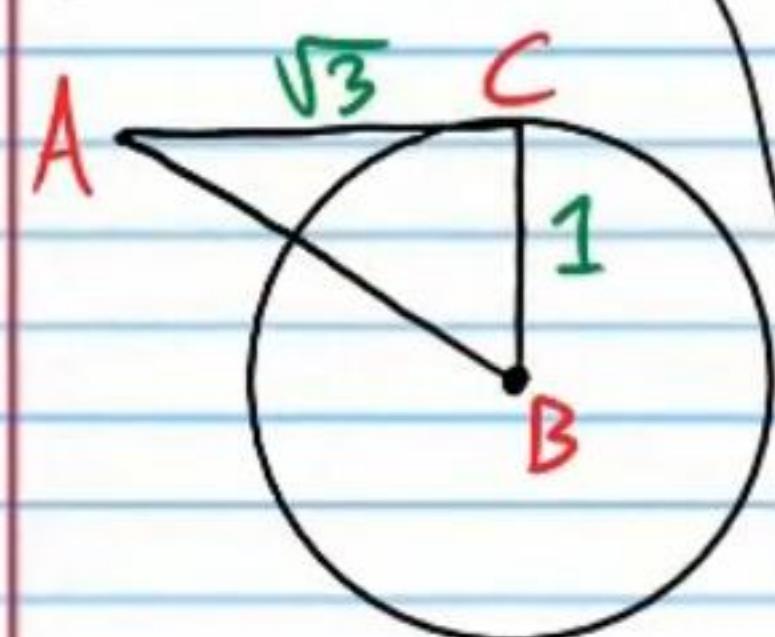
$$\boxed{\text{Yes}}$$

Pythagorean Converse Theorem: If the square of the length of the longest side of a triangle is equal to the sum of the squares of the lengths of the other sides, then the triangle is a right triangle.



If $c^2 = a^2 + b^2$, then $\triangle ABC$ is a right triangle.

⑩ $AB = 3$



$$a^2 + b^2 \stackrel{?}{=} c^2$$

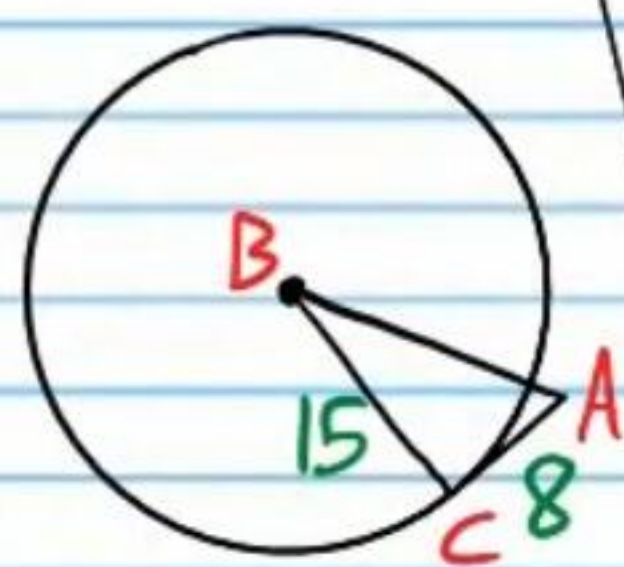
$$1^2 + (\sqrt{3})^2 \stackrel{?}{=} 3^2$$

$$1 + 3 \stackrel{?}{=} 9$$

$$4 \neq 9$$

No

⑪ $AB = 16$



$$a^2 + b^2 \stackrel{?}{=} c^2$$

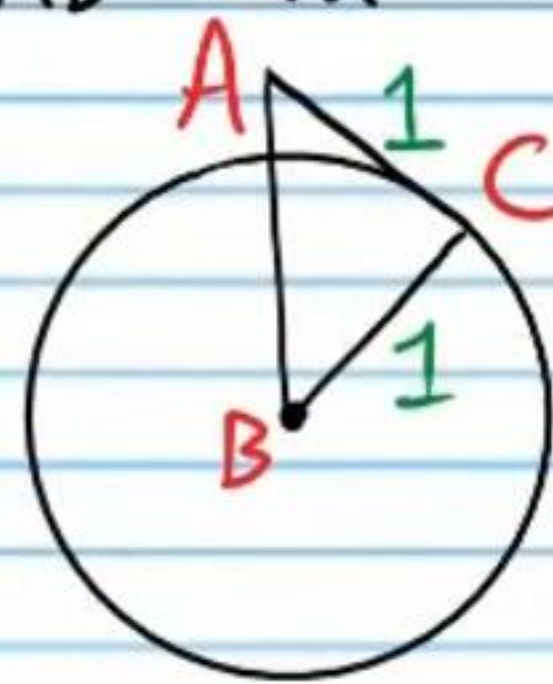
$$15^2 + 8^2 \stackrel{?}{=} 16^2$$

$$225 + 64 \stackrel{?}{=} 256$$

$$289 \neq 256$$

No

⑫ $AB = \sqrt{2}$



$$a^2 + b^2 \stackrel{?}{=} c^2$$

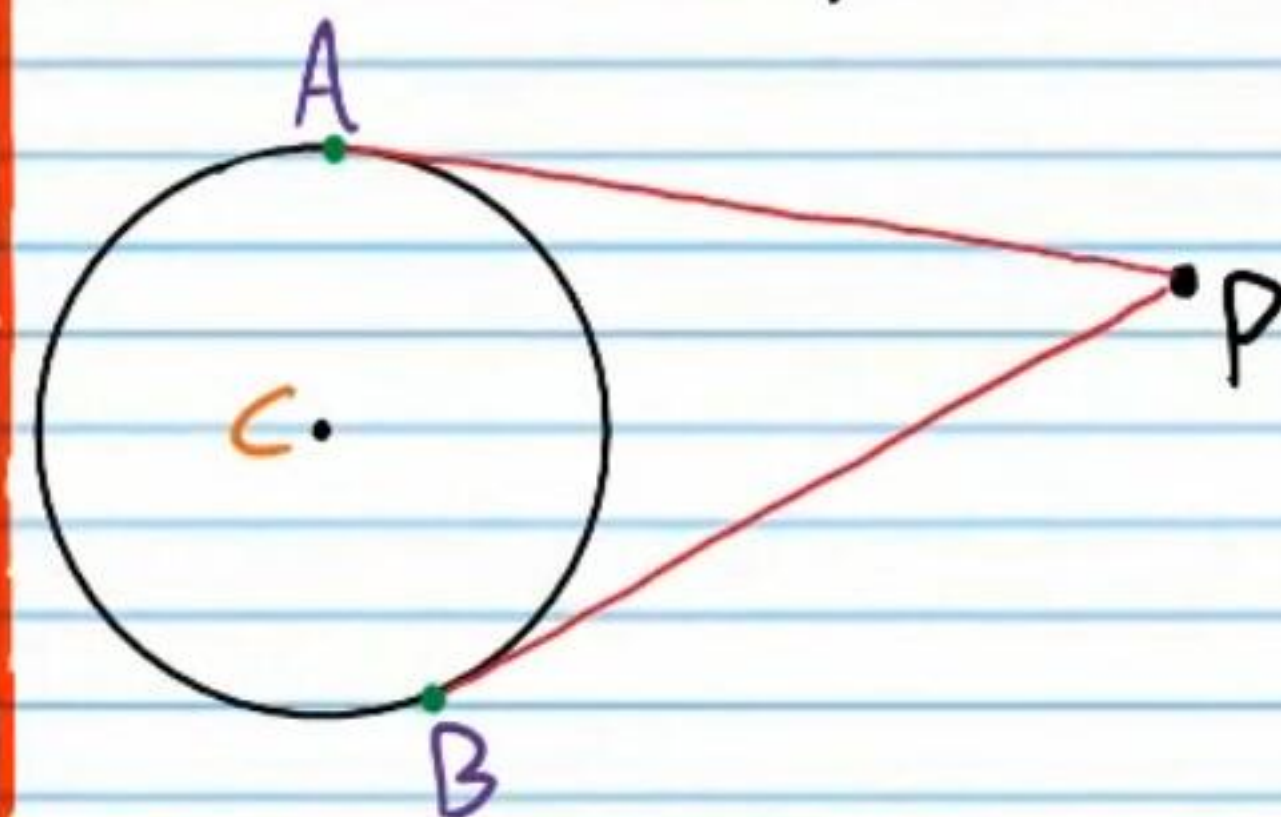
$$1^2 + 1^2 \stackrel{?}{=} (\sqrt{2})^2$$

$$1 + 1 \stackrel{?}{=} 2$$

$$2 = 2$$

Yes

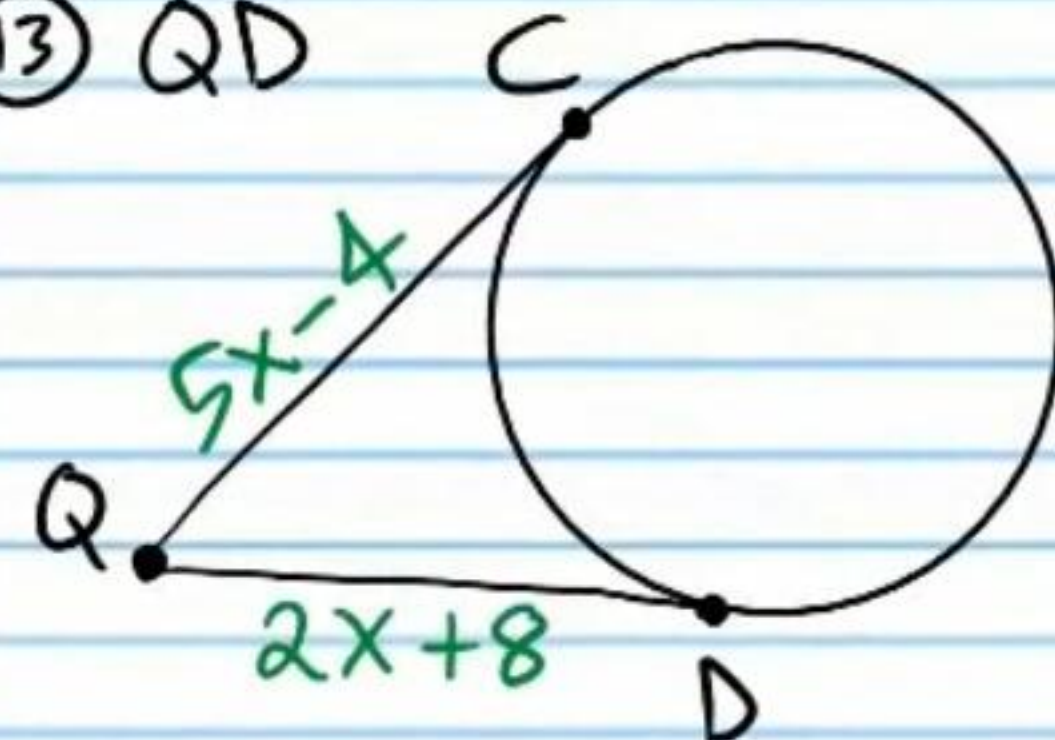
Tangents from a Common Point Theorem: If tangent segments are drawn from a common point outside a circle, then the segments are congruent.



If $\overline{AP}$ and $\overline{BP}$ are both tangent to $\odot C$, then $\overline{AP} \cong \overline{BP}$.

If C and D are points of tangency, find the following values.

⑬ QD



$$5x - 4 = 2x + 8$$

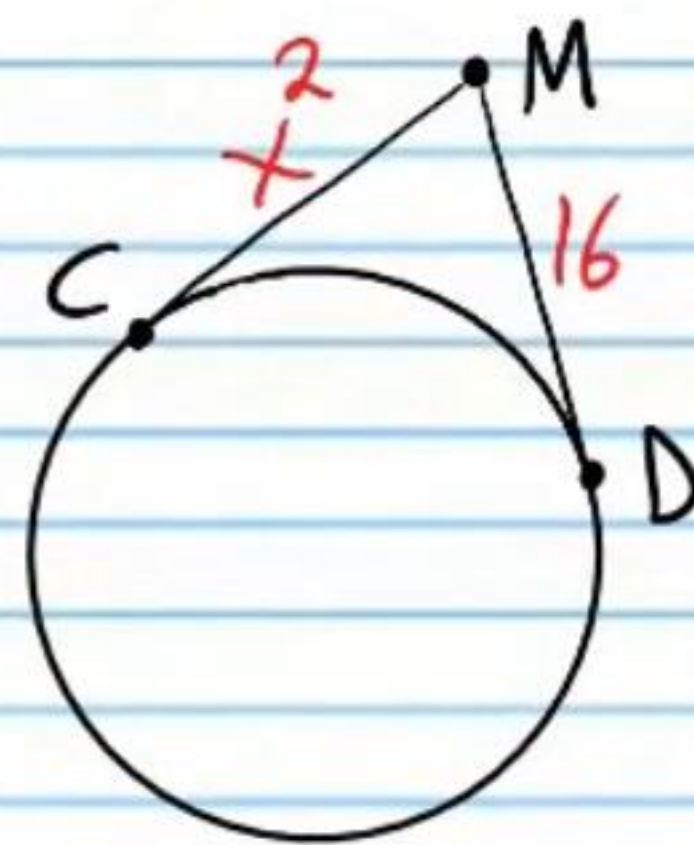
$$3x = 12$$

$$x = 4$$

$$QD = 2(4) + 8$$

$$\boxed{QD = 16}$$

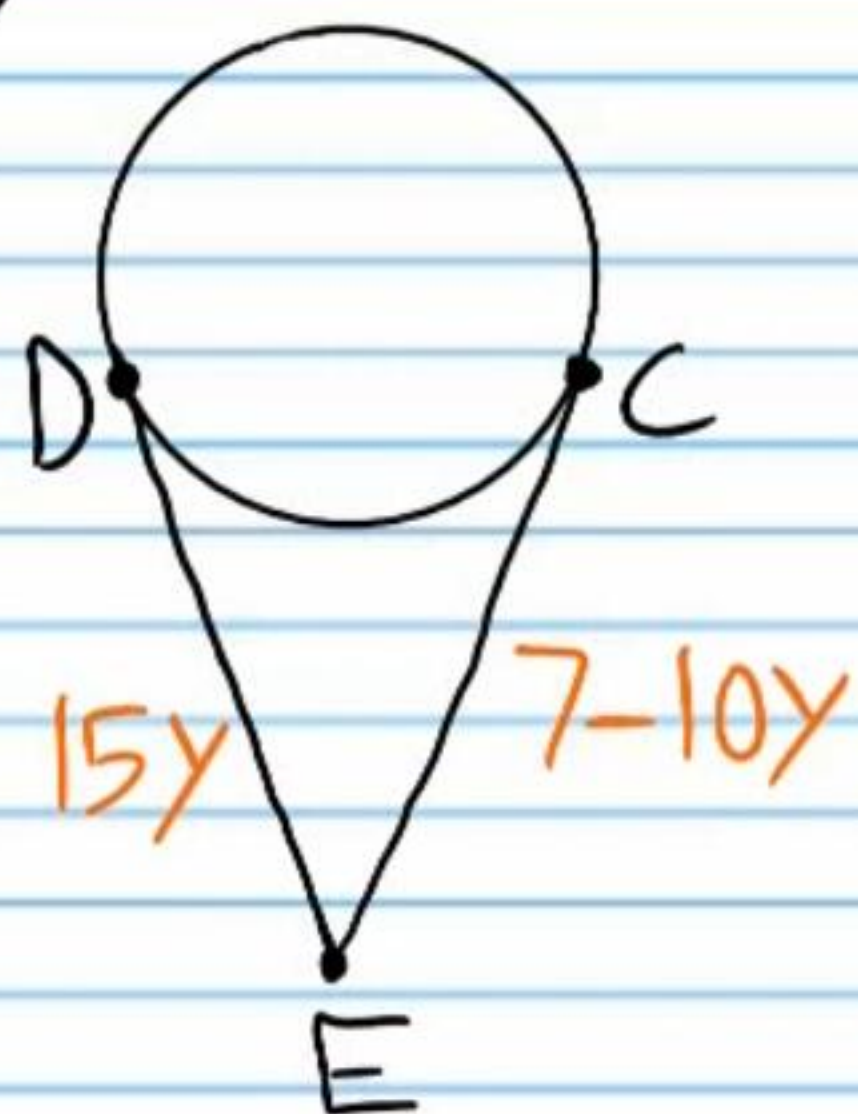
⑭ X



$$x^2 = 16$$

$$\boxed{x = \pm 4}$$

⑮ y

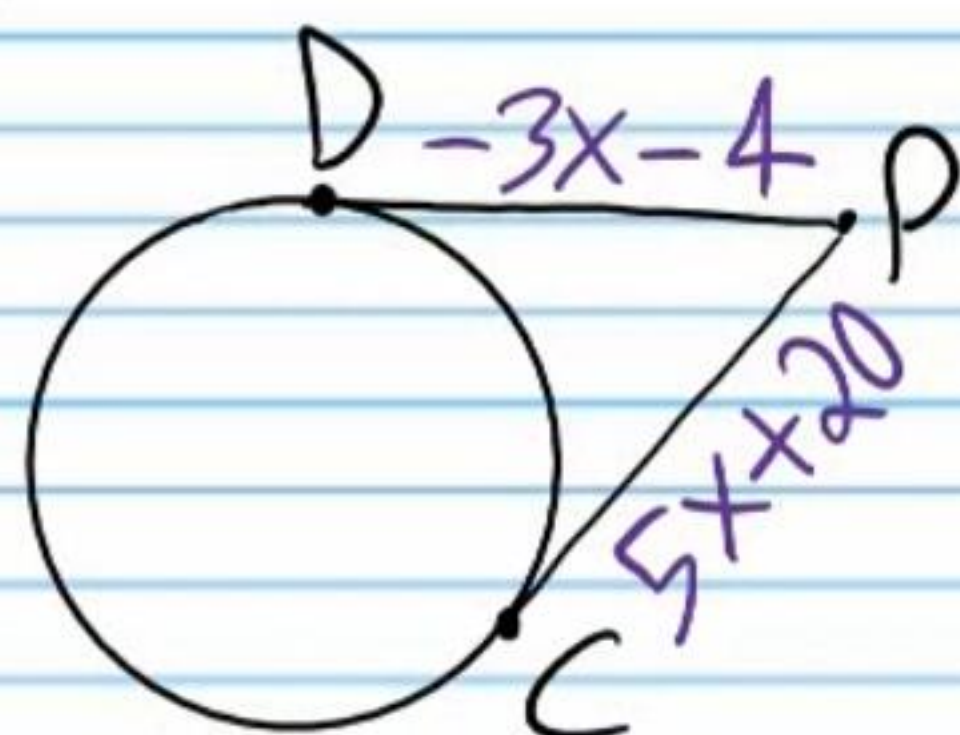


$$15y = 7 - 10y$$

$$25y = 7$$

$$y = \frac{7}{25}$$

⑯ DP



$$-3x - 4 = 5x + 20$$

$$-24 = 8x$$

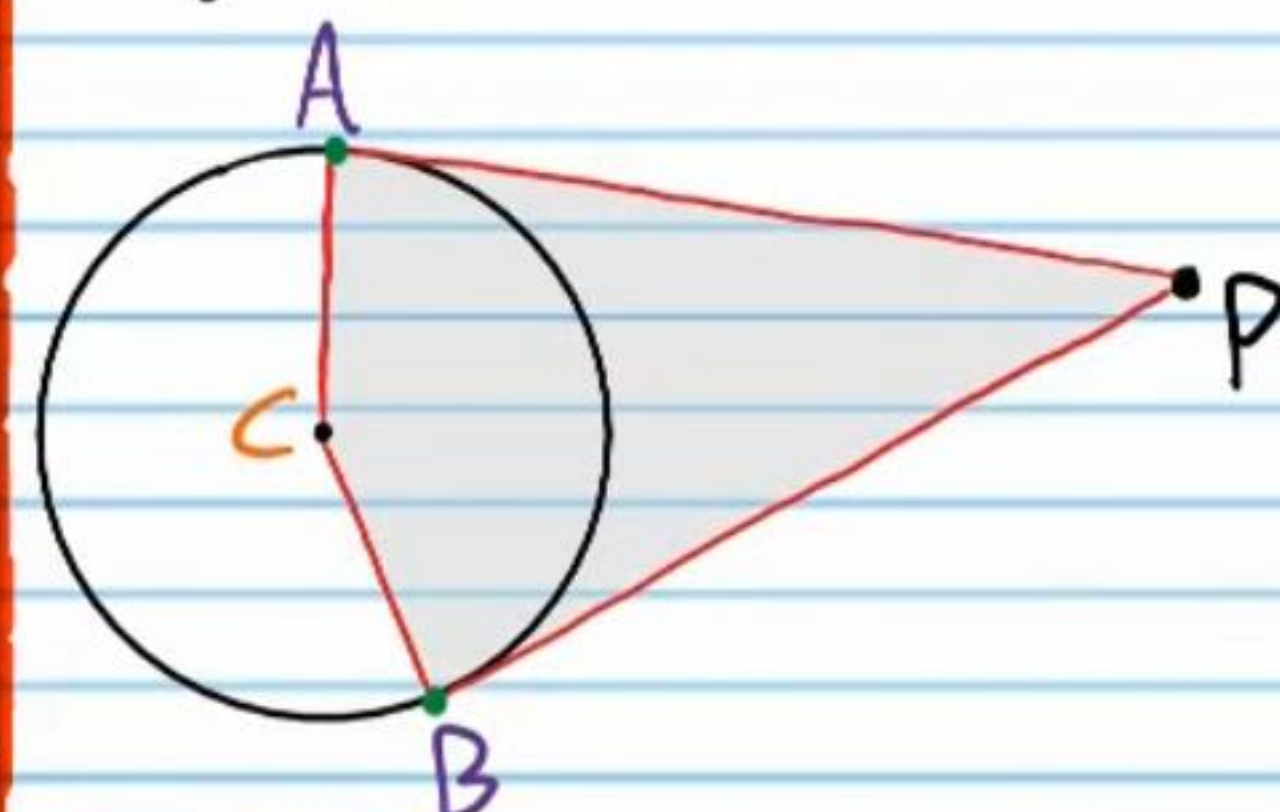
$$-3 = x$$

$$DP = -3(-3) - 4$$

$$DP = 5$$

Using Multiple Tangent Theorems

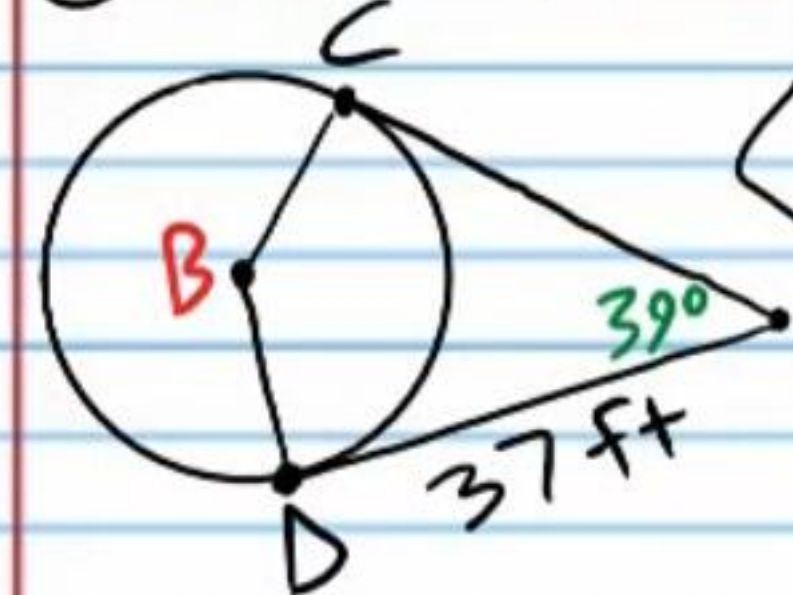
Kite formed by Tangents and Radii Theorem:
If two different tangent segments are drawn from a common point outside a circle, and two radii of the circle are drawn to the points of tangency, then the resulting quadrilateral is a kite.



ACBP is a kite.

If the segments below are radii and tangents, find the following values.

⑰ $m\angle B$ & CA.



$$m\angle A + m\angle B + m\angle C + m\angle D = 360$$

$$39 + m\angle B + 90 + 90 = 360$$

$$m\angle B = 141^\circ$$

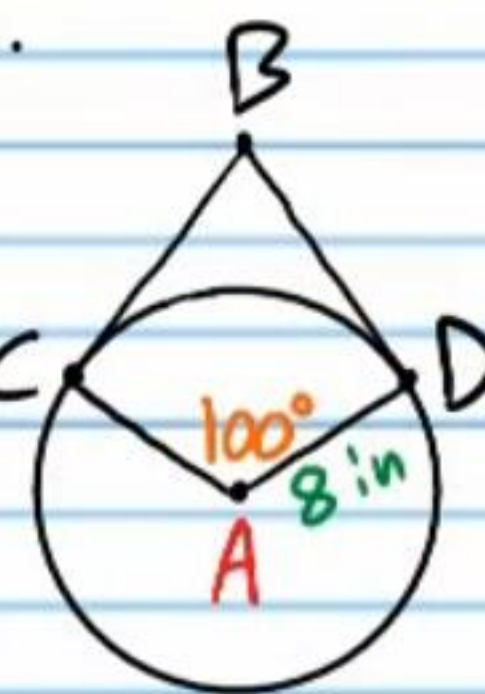
$$CA = 37 \text{ ft}$$

⑱ $m\angle B$ & CA.

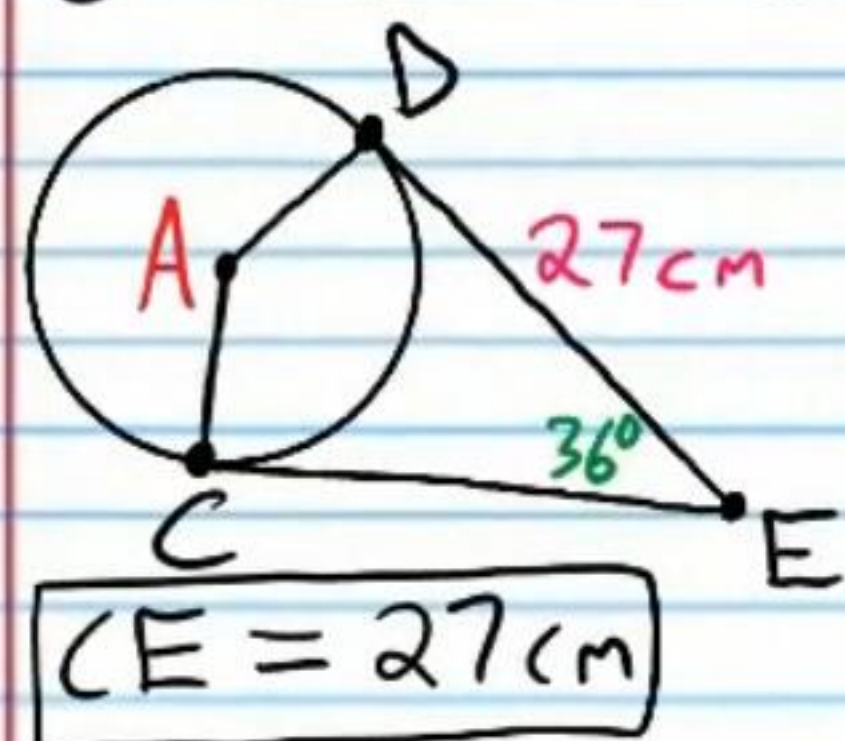
$$CA = 8 \text{ in}$$

$$m\angle B + 90 + 90 + 100 = 360$$

$$m\angle B = 80^\circ$$



①9 $m\angle DAE$ & CE .



$$90 + 90 + 36 + m\angle A = 360$$

$$m\angle A = 144^\circ$$

$$m\angle DAE = 144 \div 2$$

$$\boxed{m\angle DAE = 72^\circ}$$

②1 DL & $m\angle O$.

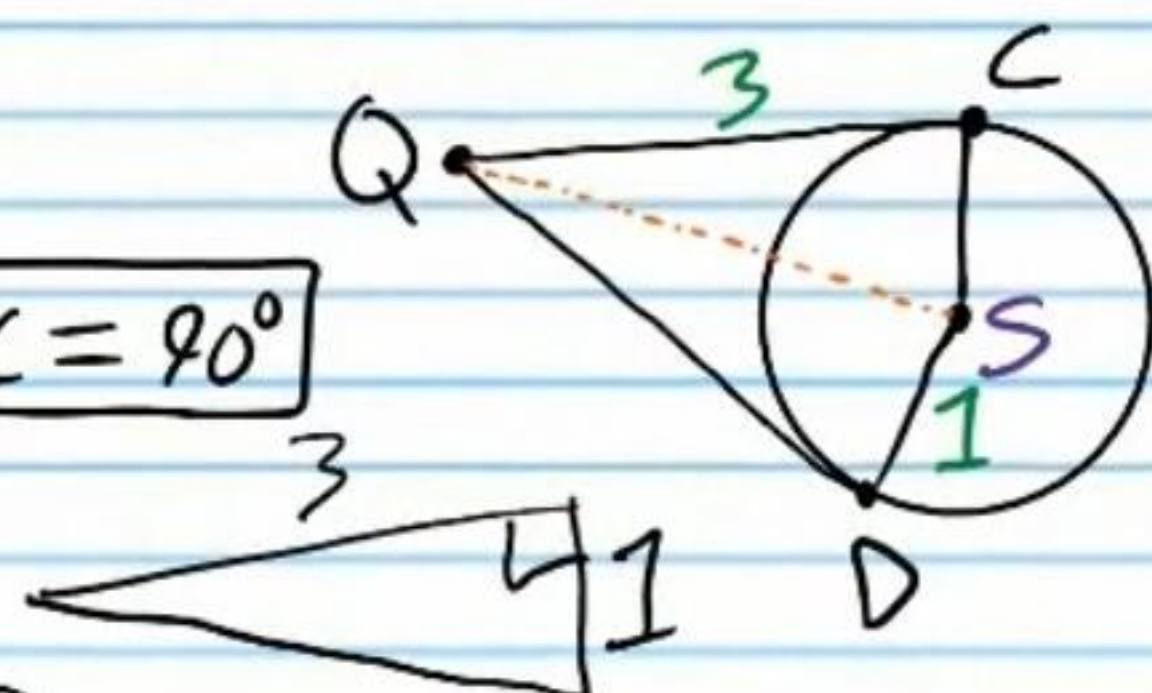


$$\boxed{DL = 10m}$$

$$90 + 90 + 109 + m\angle O = 360$$

$$\boxed{m\angle O = 71^\circ}$$

②0 QS & $m\angle C$.



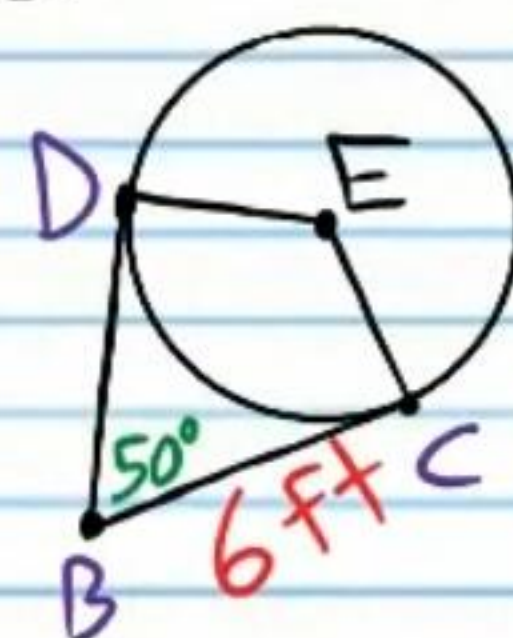
$$a^2 + b^2 = c^2$$

$$3^2 + 1^2 = c^2$$

$$10 = c^2$$

$$\boxed{QS = \sqrt{10}}$$

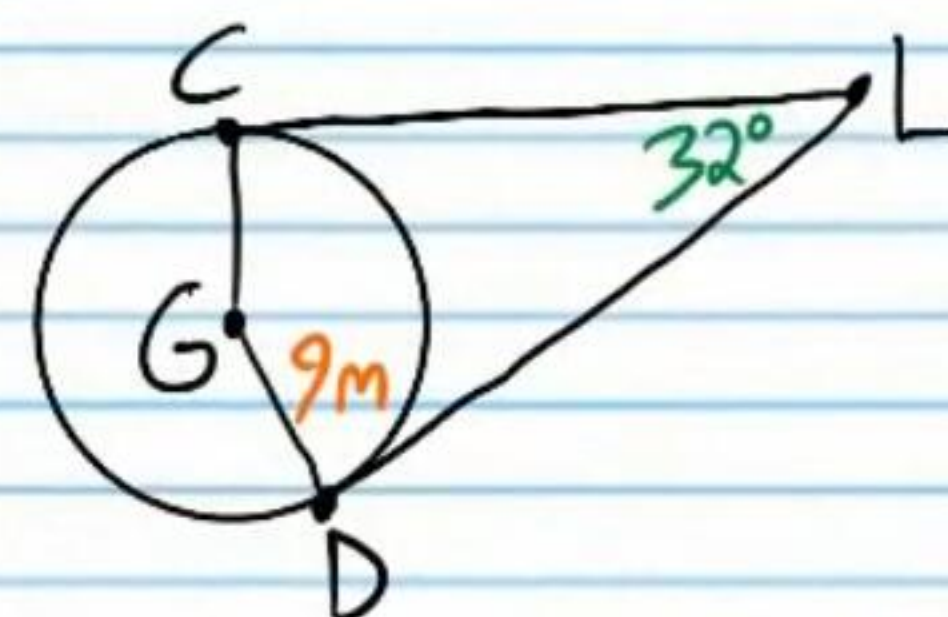
②2 BD & $m\angle E$



$$90 + 90 + 50 + m\angle E = 360$$

$$\boxed{m\angle E = 130^\circ}$$

②3 $m\angle DLG$ & CG

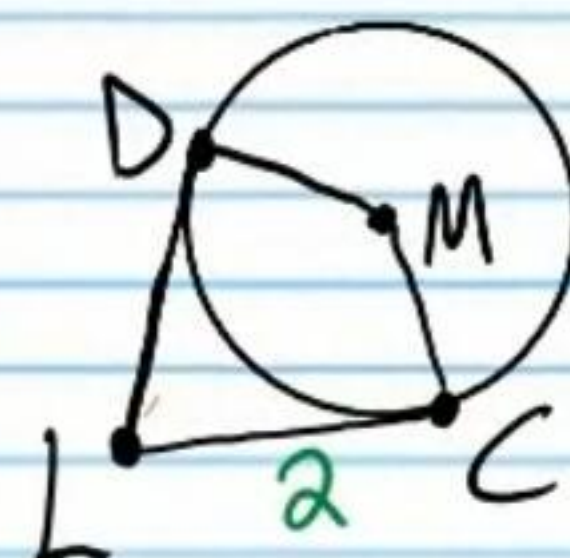


$$\boxed{CG = 9m}$$

$$m\angle DLG = 32 \div 2$$

$$\boxed{m\angle DLG = 16^\circ}$$

②4 If $LM = \sqrt{6}$, DM .



$$a^2 + b^2 = c^2$$

$$2^2 + (MC)^2 = (\sqrt{6})^2$$

$$4 + (MC)^2 = 6$$

$$(MC)^2 = 2$$

$$MC = \sqrt{2}$$

$$\boxed{DM = \sqrt{2}}$$

Arcs

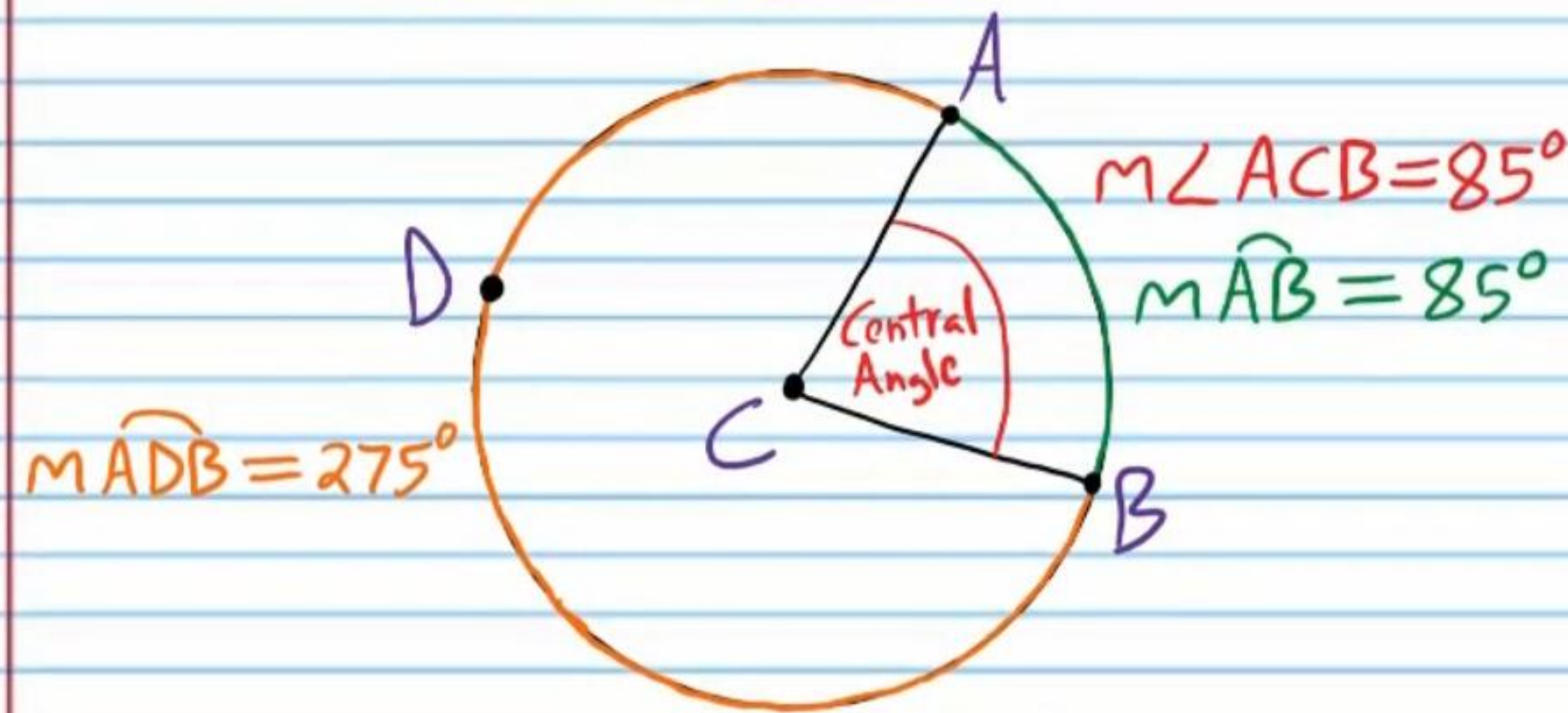
Central Angle: An angle whose vertex is on the center of a circle.

Arc Measure: A value indicating the size of a part of a circle in relation to the rest of the circle (measured in degrees). Its maximum value is 360° (representing the whole circle) and is greater than 0° .

Arc measure is defined to be equal to the measure of the corresponding central angle generated by positioning two radii at the endpoints of the arc.

Example:

$$m\angle ACB = m\widehat{AB}$$

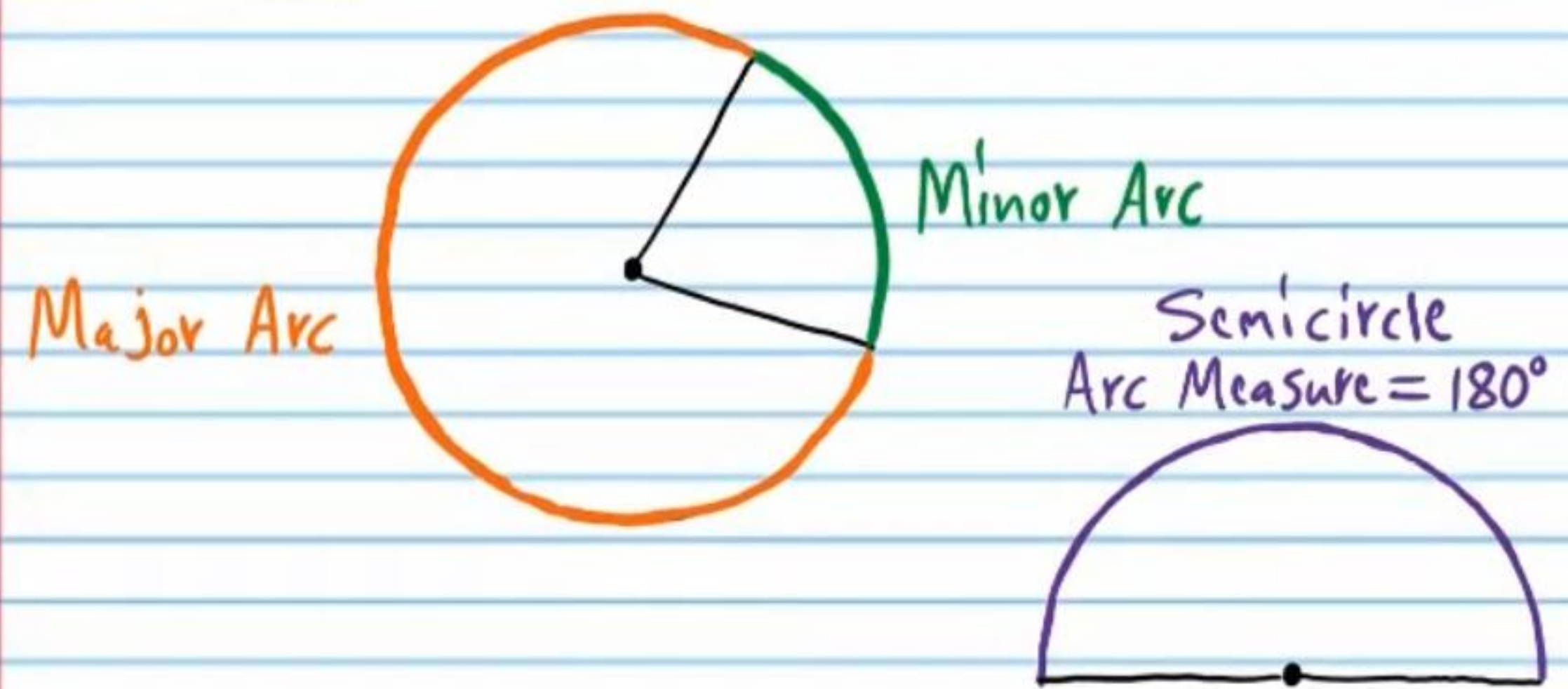


Arc Length: The actual length of an arc.
* This value is not the same as arc measure.

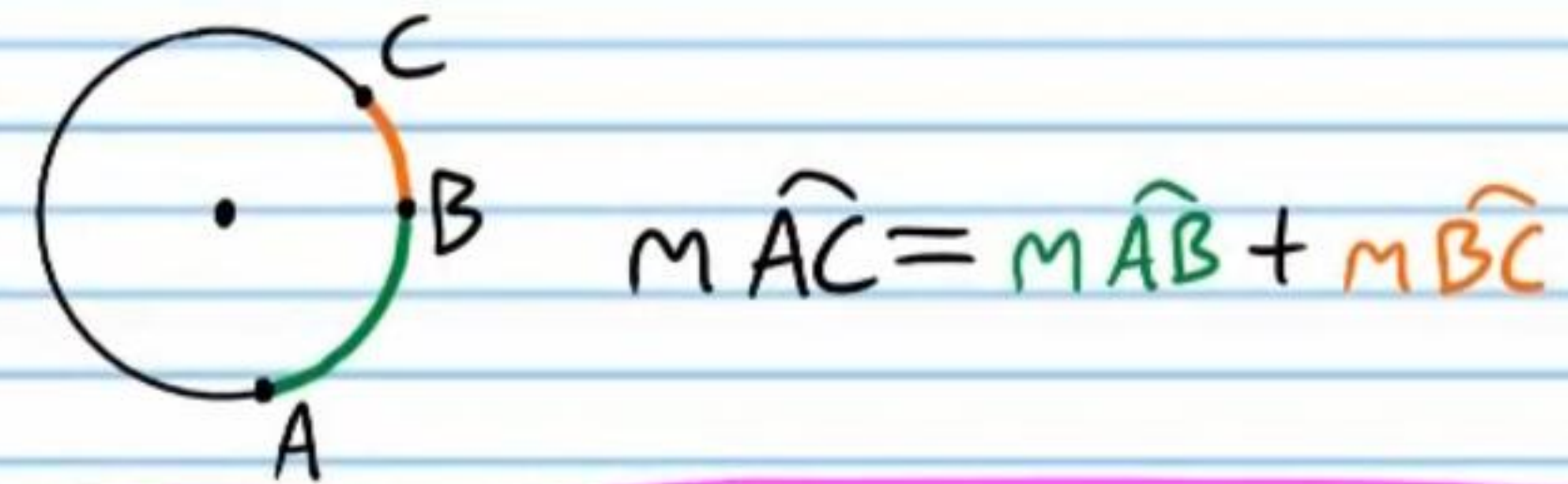
Minor Arc: An arc whose measure is less than 180° .

Major Arc: An arc whose measure is greater than 180° .

Semicircle: An arc whose measure is 180° .

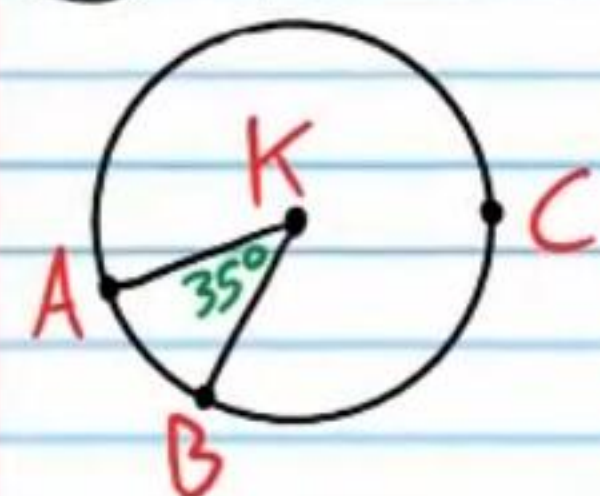


Arc Addition Postulate: The measure of an arc formed by two adjacent arcs is the sum of the measures of the two arcs.



If K is the centers of the following circles, find the values below.

25) $m\widehat{AB}$ & $m\widehat{ACB}$



$$m\widehat{AB} = 35^\circ$$

$$m\widehat{ACB} = 360 - 35$$

$$m\widehat{ACB} = 325^\circ$$

27) $m\angle HKI$ & $m\widehat{HJI}$

* $\overline{HJ}$ is a diameter.



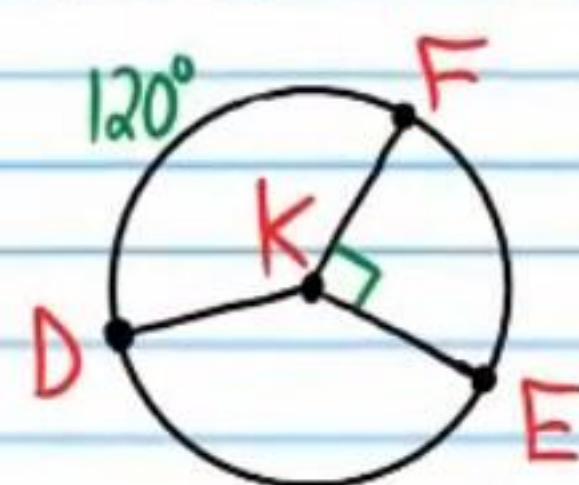
$$m\angle HKI = 180 - 118$$

$$m\angle HKI = 62^\circ$$

$$m\widehat{HJI} = 180 + 118$$

$$m\widehat{HJI} = 298^\circ$$

26) $m\angle DKE$ & $m\widehat{DFE}$



$$m\angle DKE = 360 - 90 - 120$$

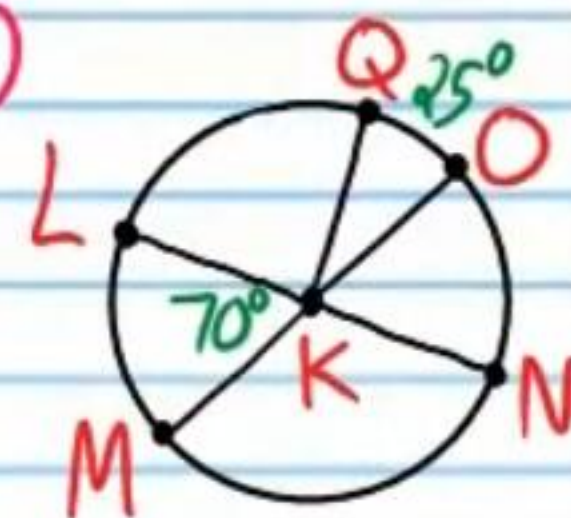
$$m\angle DKE = 150^\circ$$

$$m\widehat{DFE} = 120 + 90 = 210^\circ$$

$$m\widehat{DFE} = 210^\circ$$

28) $m\widehat{ONM}$ & $m\angle LKQ$

* $\overline{LN}$ & $\overline{MO}$ are diameters.

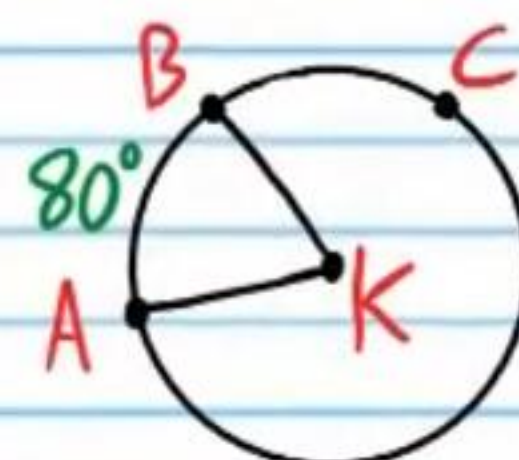


$$m\widehat{ONM} = 180^\circ$$

$$m\angle LKQ = 180 - 70 - 25$$

$$m\angle LKQ = 85^\circ$$

29) $m\angle AKB$ & $m\widehat{BCA}$



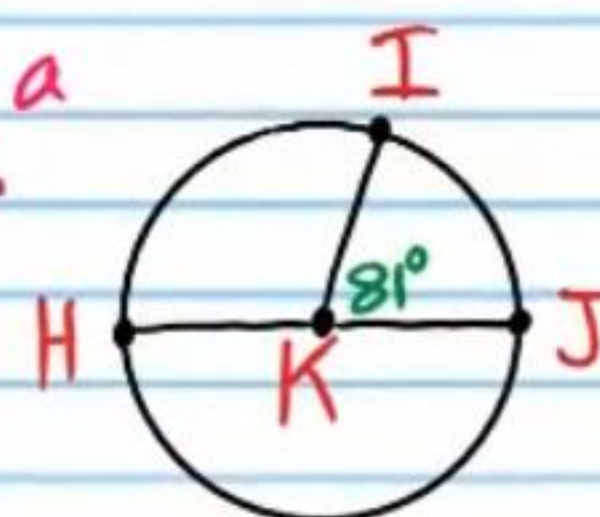
$$m\angle AKB = 80^\circ$$

$$m\widehat{BCA} = 360 - 80$$

$$m\widehat{BCA} = 280^\circ$$

31) $m\widehat{IHJ}$ & $m\angle HKI$

* $\overline{HJ}$ is a diameter.



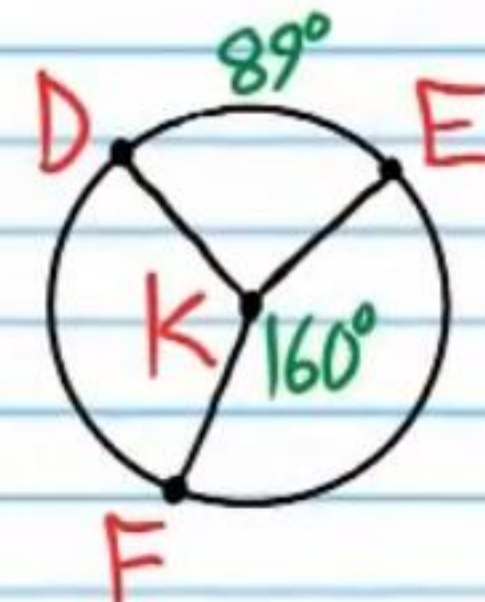
$$m\widehat{IHJ} = 360 - 81$$

$$m\widehat{IHJ} = 279^\circ$$

$$m\angle HKI = 180 - 81$$

$$m\angle HKI = 99^\circ$$

30) $m\angle DKE$ & $m\widehat{DFE}$



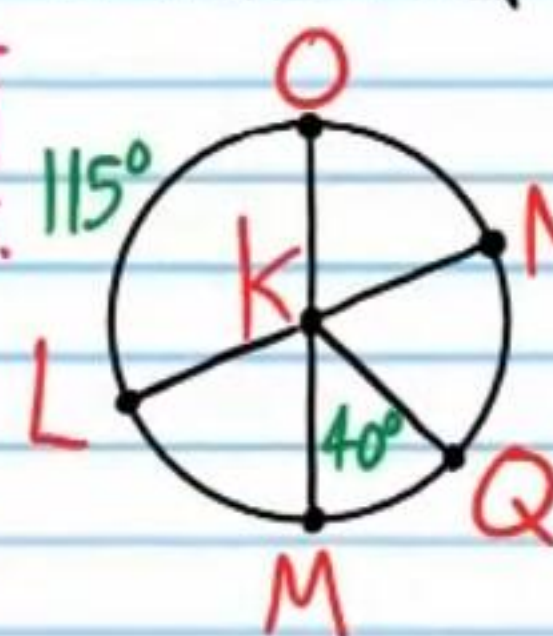
$$m\angle DKE = 89^\circ$$

$$m\widehat{DFE} = 360 - 160 - 89$$

$$m\widehat{DFE} = 111^\circ$$

32) $m\angle MKN$ & $m\widehat{LMQ}$

* $\overline{LN}$ & $\overline{OM}$ are diameters.



$$m\angle MKN = 115^\circ$$

$$m\widehat{LM} = 180 - 115 = 65^\circ$$

$$m\widehat{LMQ} = 65^\circ + 40^\circ = 105^\circ$$

$$m\widehat{LMQ} = 105^\circ$$

Circumference

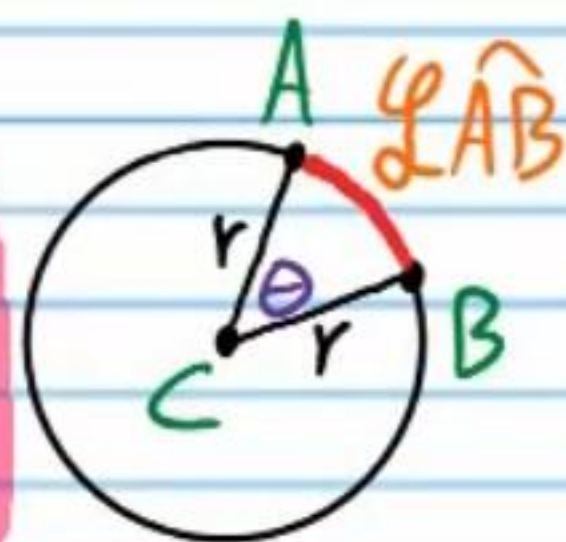
$$C = 2\pi r \text{ or } \pi D$$

"C" is circumference, "r" is radius, and "D" is diameter.

Arc Length

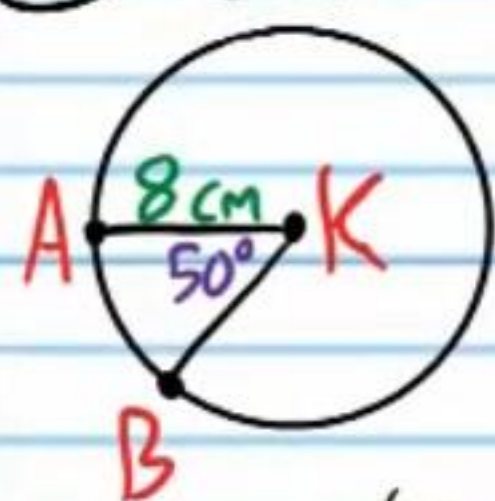
$$S = \frac{\theta}{360^\circ} (2\pi r)$$

"S" is arc length, θ is central angle, and "r" is radius.



If K is the centers of the circles below, find the lengths of the following arcs. Round to the nearest hundredth. **You must use the π button on your calculator.**

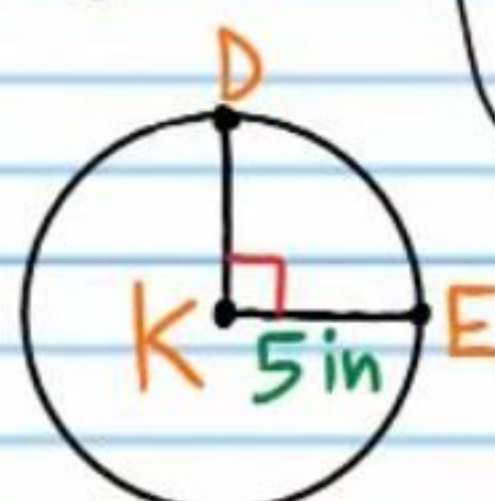
33) $\widehat{AB}$



$$\widehat{AB} = \left(\frac{50}{360}\right) (2\pi(8))$$

$$\widehat{AB} \approx 6.98 \text{ cm}$$

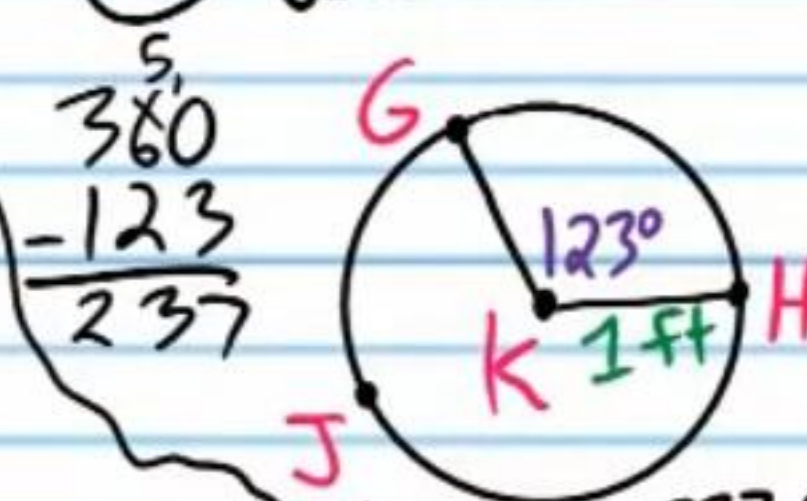
34) $\widehat{DE}$



$$\widehat{DE} = \frac{90}{360} (2\pi(5))$$

$$\widehat{DE} \approx 7.85 \text{ in}$$

35) $\widehat{GJH}$

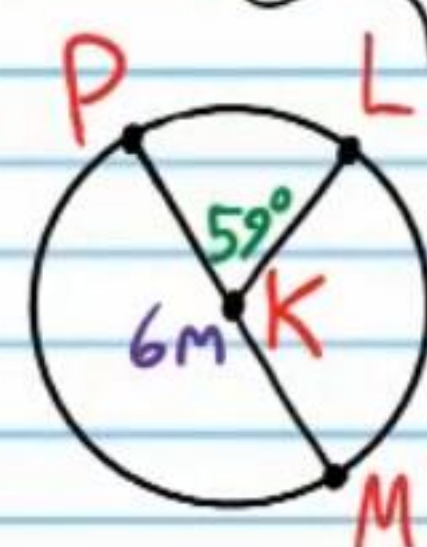


$$\widehat{GJH} = \frac{123}{360} (2\pi(1))$$

$$\widehat{GJH} \approx 4.14 \text{ ft}$$

36) $\widehat{LM}$

$\overline{PM}$ is a diameter.



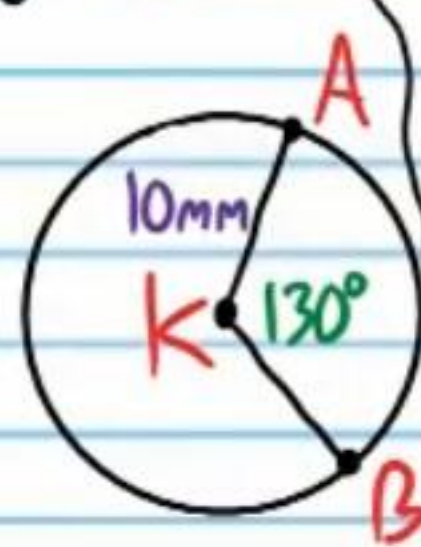
$$m\angle LKM = 180 - 59$$

$$m\angle LKM = 121^\circ$$

$$\widehat{LM} = \frac{121}{360} (2\pi(\frac{6}{2}))$$

$$\widehat{LM} \approx 6.34 \text{ m}$$

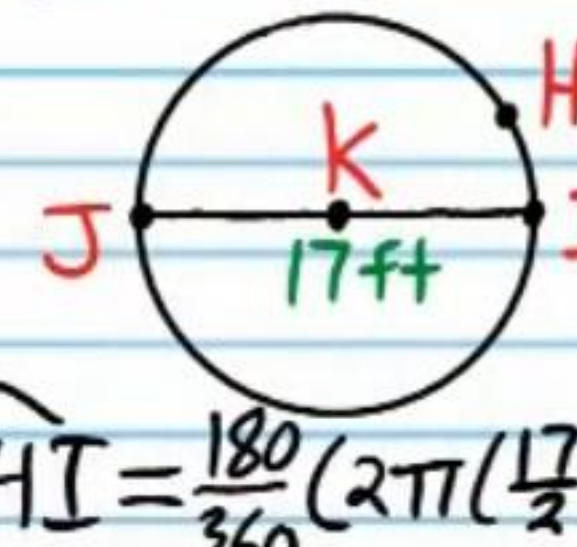
37) $\widehat{AB}$



$$\widehat{AB} = \frac{130}{360} (2\pi(10))$$

$$\widehat{AB} \approx 22.69 \text{ mm}$$

38) $\widehat{JHI}$



$$\widehat{JHI} = \frac{171}{360} (2\pi(\frac{17}{2}))$$

$$\widehat{JHI} \approx 26.7 \text{ ft}$$

39) $\widehat{DGE}$



$$m\widehat{DGE} = 360 - 43$$

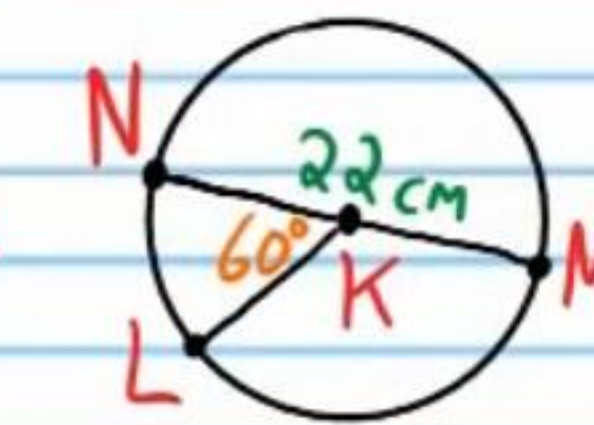
$$= 317^\circ$$

$$\widehat{DGE} = \frac{317}{360} (2\pi(9))$$

$$\widehat{DGE} \approx 49.79 \text{ in}$$

40) $\widehat{LM}$

$\overline{NM}$ is a diameter.



$$m\angle LKM = 180 - 60$$

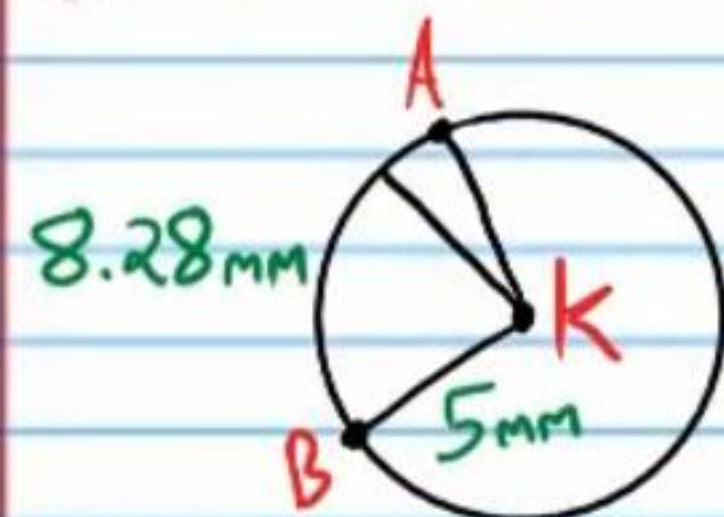
$$= 120^\circ$$

$$\widehat{LM} = \frac{120}{360} (2\pi(\frac{22}{2}))$$

$$\widehat{LM} \approx 23.04 \text{ cm}$$

If K is the centers of the circles below, find the lengths of the following arcs. Round to the nearest hundredth. **★ You must use the π button on your calculator.**

④① $m\angle AKB$



$$L\hat{A}B = \frac{\theta}{360} (2\pi r)$$

$$8.28 = \frac{\theta}{360} (2\pi(5))$$

$$8.28 = \frac{\theta}{360} (10\pi)$$

$$360 \cdot 8.28 = \frac{\theta}{360} (10\pi) \cdot 360$$

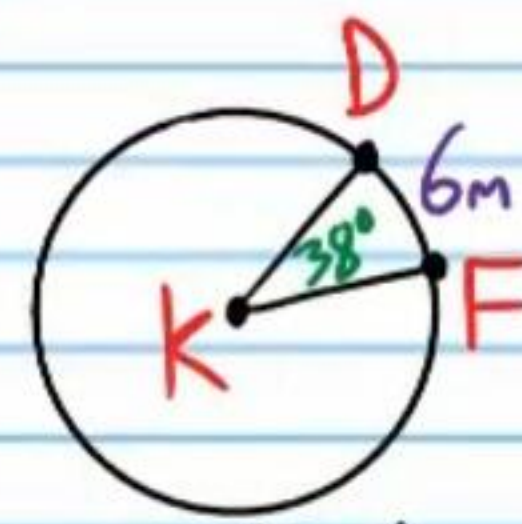
$$2980.8 = \theta (10\pi)$$

$$\frac{2980.8}{10\pi} = \frac{\theta (10\pi)}{10\pi}$$

$$\frac{2980.8}{10\pi} = \theta$$

$$94.88^\circ \approx m\angle AKB$$

④② DK



$$L\hat{D}F = \frac{\theta}{360} (2\pi r)$$

$$6 = \frac{38}{360} (2\pi r)$$

$$360 \cdot 6 = \frac{38}{360} (2\pi r) \cdot 360$$

$$360 \cdot 6 = 38 (2\pi r)$$

$$360 \cdot 6 = 38 (2\pi r)$$

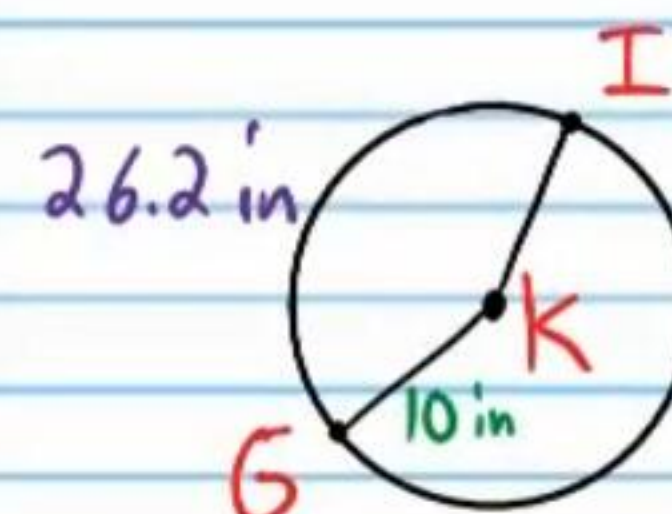
$$360 \cdot 6 = (38(2\pi)) r$$

$$\frac{360 \cdot 6}{38(2\pi)} = \frac{(38(2\pi)) r}{(38(2\pi))}$$

$$\frac{360 \cdot 6}{38(2)(\pi)} = r$$

$$9.05 m \approx DK$$

④③ $m\hat{GI}$



$$L\hat{GI} = \frac{\theta}{360} (2\pi r)$$

$$26.2 = \frac{\theta}{360} (2\pi(10))$$

$$360(26.2) = \frac{\theta}{360} (2\pi(10)) \cdot 360$$

$$360(26.2) = \theta (2\pi(10))$$

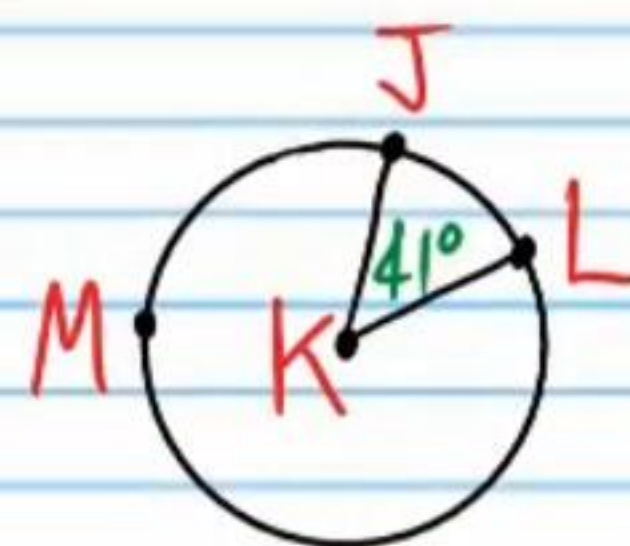
$$\frac{360(26.2)}{2\pi(10)} = \frac{\theta (2\pi(10))}{2\pi(10)}$$

$$\frac{360(26.2)}{2\pi(10)} = \theta$$

$$150.11^\circ \approx m\angle GKI$$

$$150.11^\circ \approx m\hat{GI}$$

④④ Circumference if $L\hat{JML} = 45$ ft.



$$m\hat{JML} = 360 - 41 = 319^\circ$$

$$L\hat{JML} = \frac{\theta}{360} (2\pi r)$$

$$45 = \frac{319}{360} (2\pi r)$$

$$(360)45 = \frac{319}{360} (2\pi r)(360)$$

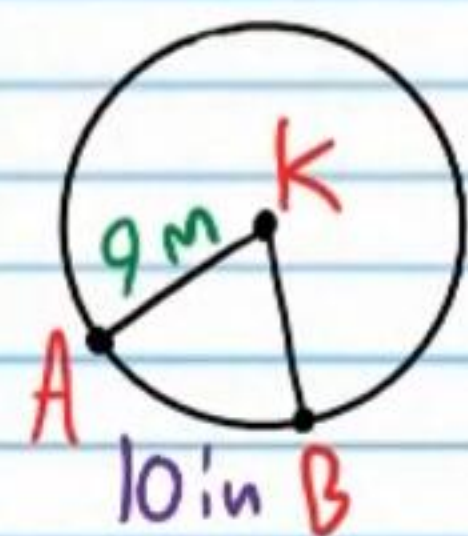
$$360(45) = 319 (2\pi r)$$

$$\frac{360(45)}{319} = \frac{319(2\pi r)}{319}$$

$$\frac{360(45)}{319} = 2\pi r$$

$$50.78 ft \approx 2\pi r$$

45) $m\hat{A}B$ if



$$\angle A\hat{B} = \frac{\theta}{360} (2\pi r)$$

$$10 = \frac{\theta}{360} (2\pi(9))$$

$$(360)10 = \frac{\theta}{360} (2\pi(9))(360)$$

$$360(10) = \theta (2\pi(9))$$

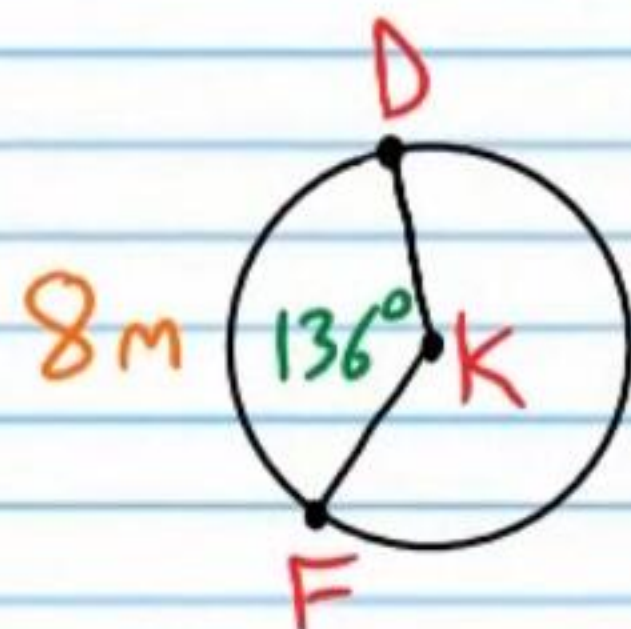
$$\frac{360(10)}{2\pi(9)} = \frac{\theta (2\pi(9))}{2\pi(9)}$$

$$\frac{360(10)}{2\pi(9)} = \theta$$

$$63.66^\circ \approx m\angle AKB$$

$$\boxed{63.66^\circ \approx m\hat{A}B}$$

46) Diameter



$$\angle D\hat{F} = \frac{\theta}{360} (2\pi r)$$

$$8 = \frac{136}{360} (2\pi r)$$

$$(360)8 = \frac{136}{360} (2\pi r)(360)$$

$$(360)8 = 136 (2\pi r)$$

$$(360)8 = (136(2\pi))r$$

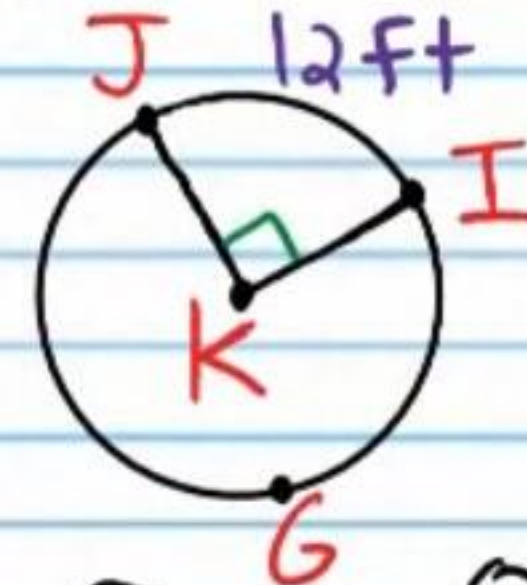
$$\frac{(360)8}{136(2\pi)} = \frac{(136(2\pi))r}{136(2\pi)}$$

$$\frac{360(8)}{136(2\pi)} = r$$

$$2 \left(\frac{360(8)}{136(2\pi)} \right) = D$$

$$\boxed{6.74m \approx D}$$

47) $\angle JGI$



$$\angle J\hat{I} = \frac{\theta}{360} (2\pi r)$$

$$12 = \frac{90}{360} (2\pi r)$$

$$(360)12 = \frac{90}{360} (2\pi r)(360)$$

$$(360)(12) = 90 (2\pi r)$$

$$\frac{360(12)}{90} = \frac{90 (2\pi r)}{90}$$

$$\frac{360(12)}{90} = 2\pi r$$

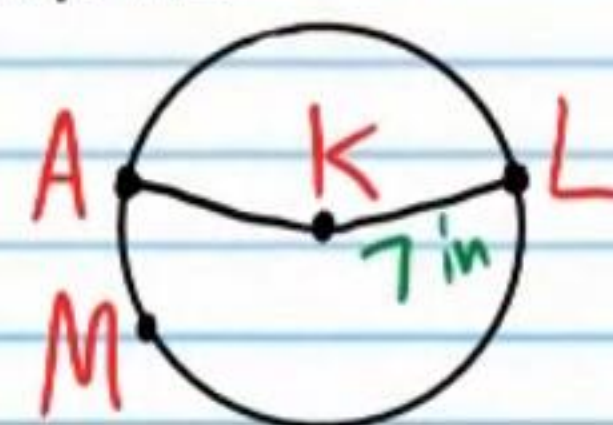
$$m\hat{JGI} = 360 - 90 = 270^\circ$$

$$\angle J\hat{GI} = \frac{\theta}{360} (2\pi r)$$

$$\angle J\hat{GI} = \frac{270}{360} \left(\frac{360(12)}{90} \right)$$

$$\boxed{\angle JGI \approx 36^\circ}$$

48) $m\angle AKL$ if
 $\angle AML = 27.5^\circ$



$$\angle A\hat{M}L = \frac{\theta}{360} (2\pi r)$$

$$27.5 = \frac{\theta}{360} (2\pi(7))$$

$$(360)27.5 = \frac{\theta}{360} (2\pi(7))(360)$$

$$360(27.5) = \theta (2\pi(7))$$

$$\frac{360(27.5)}{2\pi(7)} = \frac{\theta (2\pi(7))}{2\pi(7)}$$

$$\frac{360(27.5)}{2\pi(7)} = \theta$$

$$225.0905623 \approx \theta$$

$$m\angle AKL = 360 - \theta$$

$$\boxed{m\angle AKL \approx 134.91^\circ}$$